SOC ANALYST HOME LAB REPORT

Wazuh SIEM • Windows Security Event Monitoring • Alert Investigation

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Wazuh SIEM-Based Windows Security Monitoring and Alert Investigation

Project Type: Cybersecurity Practical Project

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Course: Cybersecurity / SOC Analyst Training

Tools Used: Wazuh SIEM, Docker Desktop, Wazuh Agent, Windows Security Logs

1. Project Overview:

This project focuses on building a Security Operations Center (SOC) home lab using Wazuh SIEM to monitor and investigate Windows security events.

The lab uses Docker-based Wazuh services and a Windows endpoint running the Wazuh agent. The Wazuh dashboard is used to review security events, examine authentication-related activity, inspect alerts, and understand the basic security event triage process.

The project provides practical exposure to Security Information and Event Management (SIEM), endpoint monitoring, event analysis, and incident investigation documentation.

Further validation of Sysmon telemetry and additional detection scenarios is planned as part of the ongoing project.

2. Intruduction:

A Security Operations Center (SOC) is responsible for monitoring systems, identifying suspicious activities, investigating security alerts, and helping organizations respond to security incidents.

Security Information and Event Management (SIEM) solutions collect and analyze logs from different systems to help security analysts identify potentially important events.

Wazuh is an open-source security monitoring platform that can collect endpoint telemetry, analyze security events, and display alerts through a dashboard.

In this project, Wazuh is used to monitor a Windows endpoint and provide a practical environment for learning security event analysis and alert investigation.

3. Projective Objectives:

The main objectives of this project are:

1. To deploy a Wazuh SIEM environment using Docker.
2. To connect and monitor a Windows endpoint using the Wazuh agent.
3. To view and filter security events through the Wazuh dashboard.
4. To review authentication-related events and security alerts.
5. To inspect event details and understand the basic alert triage process.
6. To document investigation findings and identify areas for further improvement.
7. To validate Sysmon event collection and additional detection scenarios as future milestones.

4. Tools and Technologies:

tools	**Project me use**

Wazuh	Security events aur alerts monitor karne ke liye
Wazuh Agent	Windows endpoint se telemetry collect krne ke liye
Docker Desktop	Wazuh service ke containers run karne ke liye
Windows 11	Monitored endpoint aur security event source
Windows Security Logs	Authentication

5. Lab Environment:

Component	Details
Host Operating System	Windows 11
SIEM	Wazuh
Deploymentt Method	Docker Desktop
Monitored Agent	OM-Windows-Lab
Agent ID	001
DAshboard	Wazuh Threat hunting
Monitoring Focus	Windows events

Is flow ka meaning:

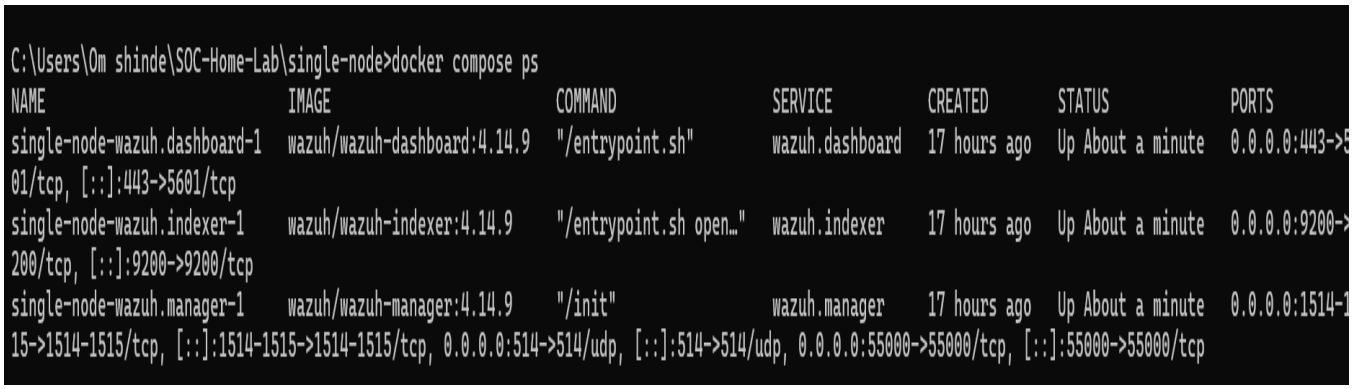
1.Windows endpoint par activity hoti hai.
2.Wazuh agent configured events collect karta hai.
3.Wazuh manager events ko process karta hai.
4.Wazuh indexer searchable data store karta hai.
5.Dashboard par analyst events aur alerts review karta hai.

6. Implementation and Configurtion:

6.1 Wazuh Deployment

Wazuh was deployed using its Docker-based single-node environment. The deployment includes the Wazuh manager, Wazuh indexer, and Wazuh dashboard services.

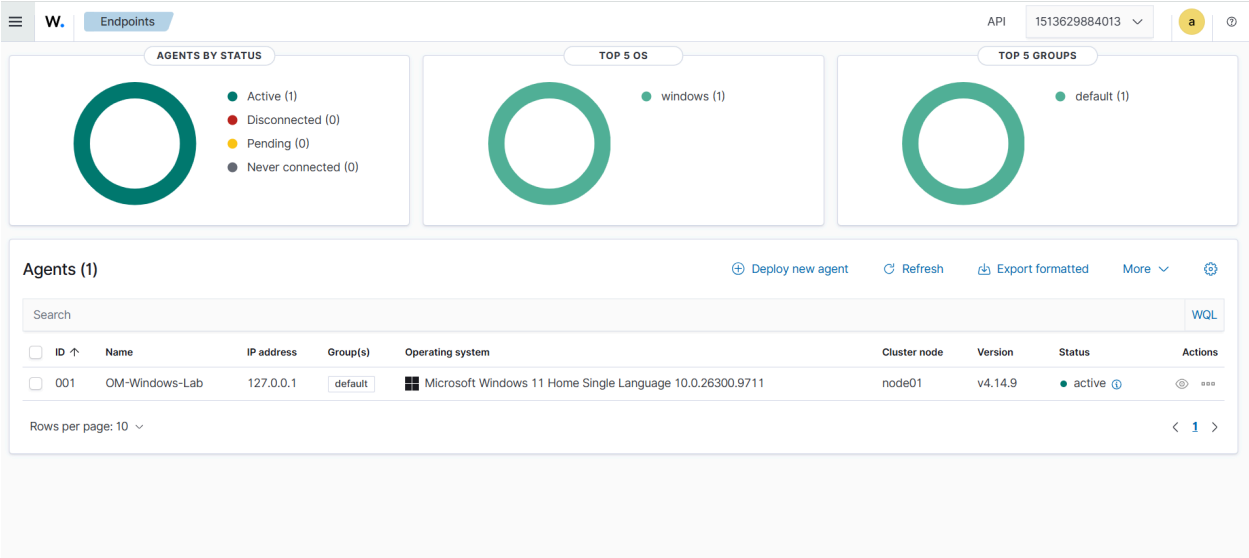
The Docker container status was checked to confirm that the Wazuh services were running. The dashboard was then accessed locally for security monitoring and event review**.**



6.2 Windows Agent Monitoring:

The Windows endpoint was configured with the Wazuh agent. The agent was named OM-Windows-Lab and appeared in the Wazuh dashboard with agent ID 001.

The dashboard was used to review events associated with this endpoint. Individual event documents were opened to inspect available event fields and understand the information used during security investigations



7. Practical Experiments:

Practical 1 — Wazuh Dashboard Event Monitoring

Objective: Windows endpoint ke collected events dashboard mein review karna.

Procedure:

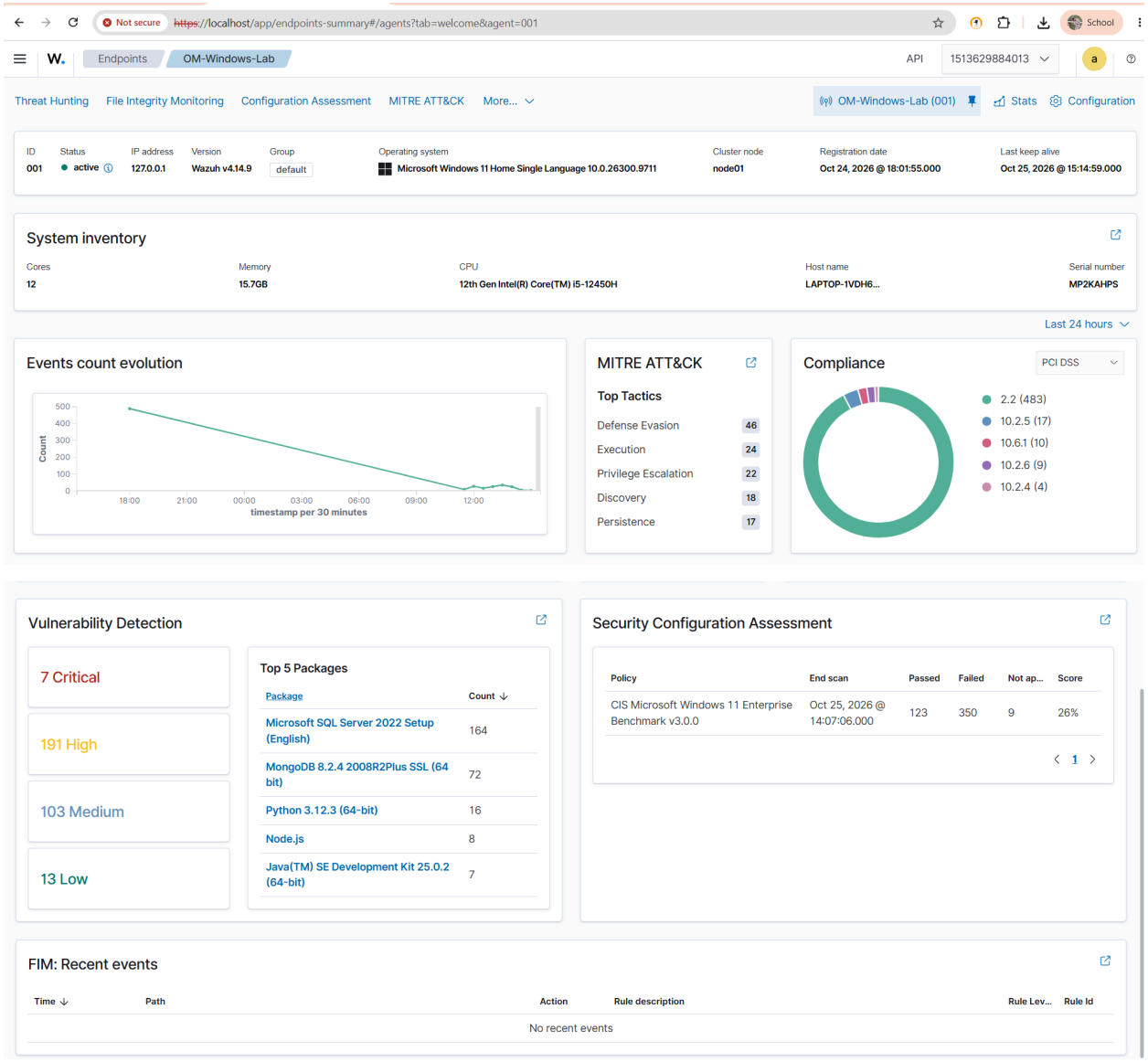
1.Wazuh dashboard open kiya.
2.OM-Windows-Lab (001) agent select kiya.
3.Threat Hunting ke Events section mein gaye.

4.Displayed time range ke events review kiye.
5.individual event ko open karke available fields check kiye.

Observation: Dashboard par Windows endpoint se associated events visible the.

Result: Event monitoring aur basic event inspection successfully demonstrate hua.

Screenshot:



Practical 2 — Authentication Event Review.

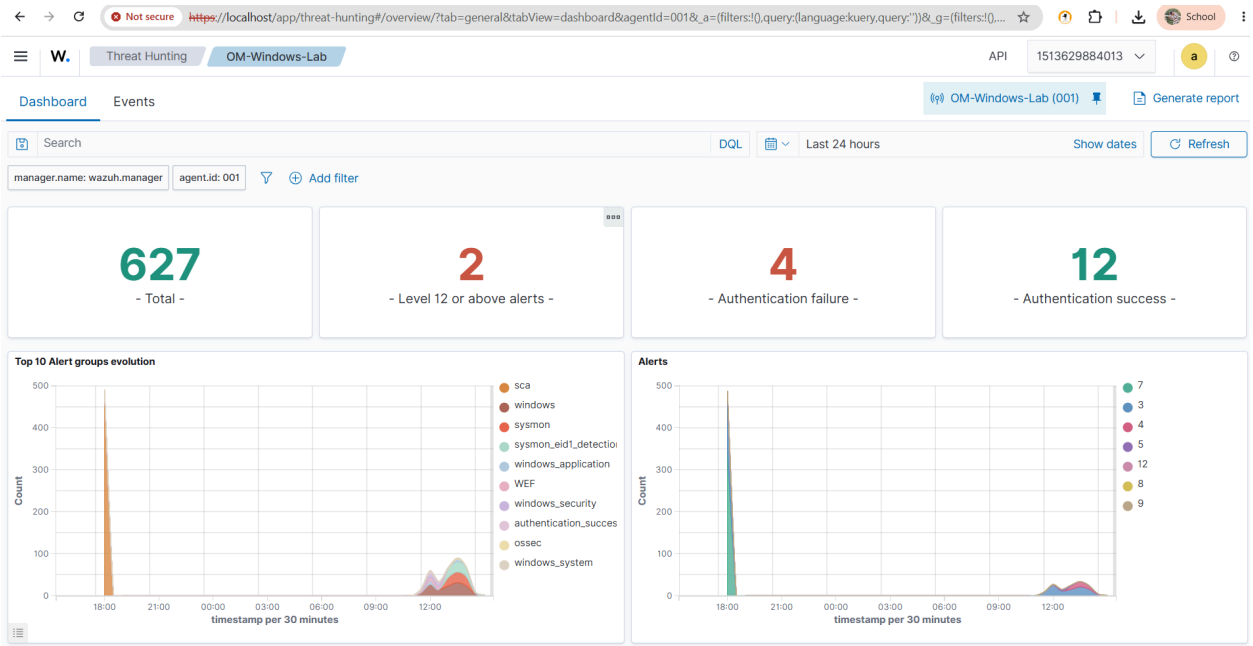
Objective: Authentication-related events ko identify aur inspect karna.

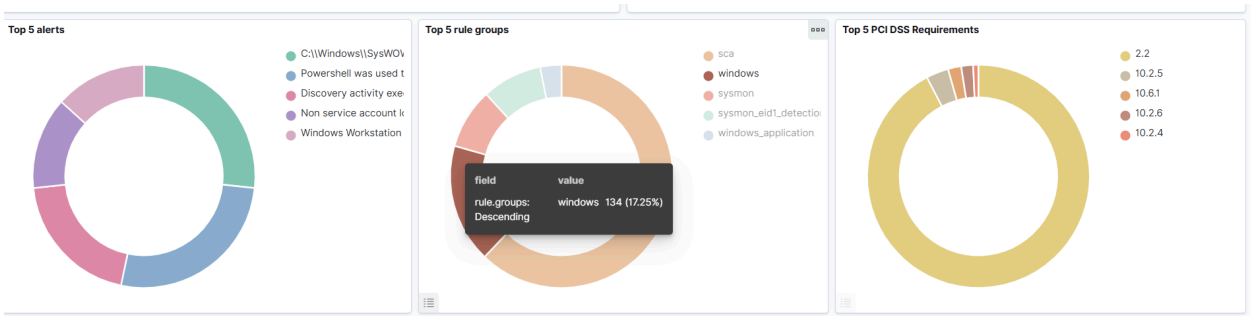
Procedure:
1.Wazuh dashboard par relevant time range select kiya.
2.Authentication-related alerts review kiye.
3.Ek event document open kiya.
4.Available fields jaise agent.name, data.win.eventdata.logonType, data.win.eventdata.processName, aur status/substatus inspect kiye.

Observation:
Screenshot mein authentication-related data dikh raha tha. Ek event mein process name C:\Windows\System32\svchost.exe aur logon type 2 bhi visible tha.

Result:
Authentication event fields ko read karne aur unka investigation mein kya role hai, ye samajhne ka practical experience mila.

Screenshots:





8. Alert Investigantion:

A. Objective:

Windows endpoint par failed login attempts ko Wazuh SIEM ke through monitor aur investigate karna.

B. Methodology:

1.Windows Security events ko Wazuh dashboard mein open kiya.
2.Agent OM-Windows-Lab select kiya.
3.Event ID 4625 ke events filter kiye.
4.Rule ID, severity, timestamp, source IP aur authentication details examine ki.
5.Multiple failed-login events ko identify kiya.

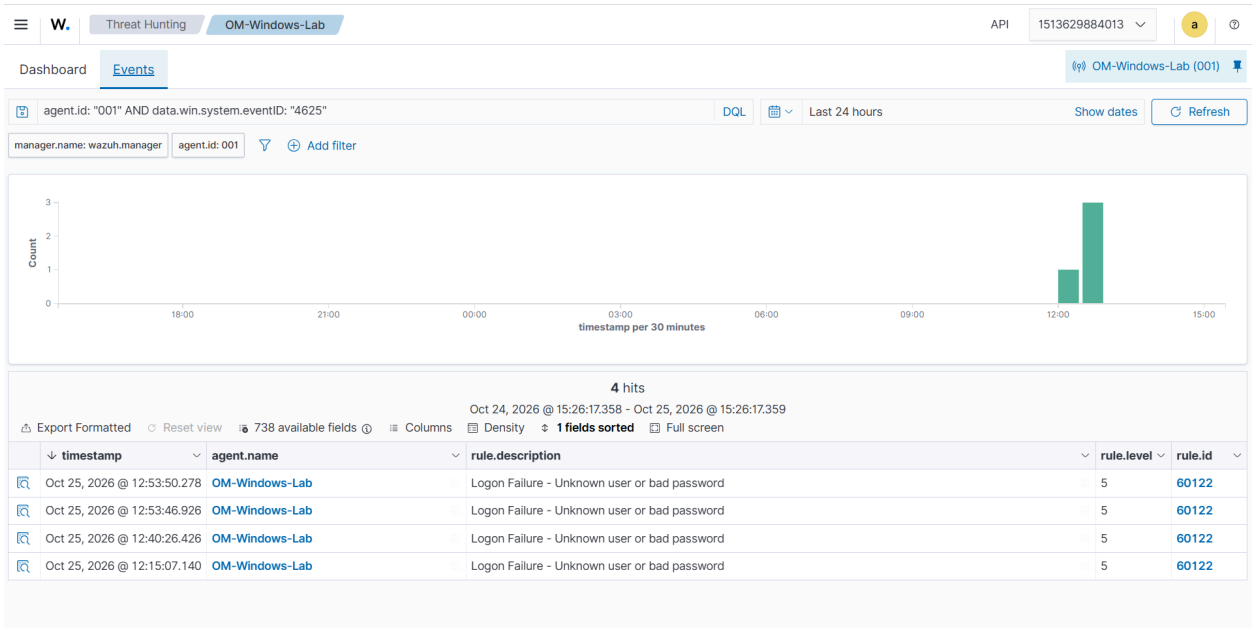
C. Findings:

Wazuh ne Windows Security Event ID 4625 ko detect kiya. Rule ID 60122 ne event ko Logon Failure - Unknown user or bad password ke roop mein classify kiya. Selected time range mein chaar matching events mile.

D. Conclusion:

Wazuh SIEM successfully received and classified Windows failed-logon events. Ye investigation authentication monitoring aur initial alert triage demonstrate karti hai.

ScreenShots:



Document Details

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Table | JSON

t _index	wazuh-alerts-4.x-2026.10.25
t agent.id	001
t agent.ip	127.0.0.1
t agent.name	OM-Windows-Lab
t data.win.eventdata.authenticationPackageName	Negotiate
t data.win.eventdata.failureReason	%%2313
t data.win.eventdata.ipAddress	127.0.0.1
t data.win.eventdata.ipPort	0
t data.win.eventdata.keyLength	0
t data.win.eventdata.logonProcessName	User32
t data.win.eventdata.logonType	2
t data.win.eventdata.processId	0x954
t data.win.eventdata.processName	C:\\Windows\\System32\\svchost.exe
t data.win.eventdata.status	0xc000006d
t data.win.eventdata.subStatus	0xc000006a
t data.win.eventdata.subjectDomainName	WORKGROUP
t data.win.eventdata.subjectLogonId	0x3e7

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t	data.win.eventdata.subjectUserName	LAPTOP-1VDH61EB$
t	data.win.eventdata.subjectUserSid	S-1-5-18
t	data.win.eventdata.targetDomainName	LAPTOP-1VDH61EB
t	data.win.eventdata.targetUserName	Om shinde
t	data.win.eventdata.targetUserSid	S-1-0-0
t	data.win.eventdata.workstationName	LAPTOP-1VDH61EB
t	data.win.system.channel	Security
t	data.win.system.computer	LAPTOP-1VDH61EB
t	data.win.system.eventID	4625
t	data.win.system.eventRecordID	701343
t	data.win.system.keywords	0x8010000000000000
t	data.win.system.level	0
t	data.win.system.message	"An account failed to log on. Subject: Security ID: S-1-5-18 Account Name: LAPTOP-1VDH61EB$ Account Domain: WORKGROUP Logon ID: 0x3E7
t	data.win.system.opcode	0
t	data.win.system.processID	1240
t	data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}

Document Details

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t	data.win.system.version	0
t	decoder.name	windows_eventchannel
t	id	1792913030.170424
t	input.type	log
t	location	EventChannel
t	manager.name	wazuh.manager
t	rule.description	Logon Failure - Unknown user or bad password
#	rule.firedtimes	3
t	rule.gdpr	IV_35.7.d, IV_32.2
t	rule.gpg13	7.1
t	rule.groups	windows, windows_security, authentication_failed
t	rule.hipaa	164.312.b
t	rule.id	60122
#	rule.level	5
◕	rule.mail	false
t	rule.mitre.id	T1531
t	rule.mitre.tactic	Impact
t	rule.mitre.technique	Account Access Removal
t	rule.nist_800_53	AU.14, AC.7
t	rule.pci_dss	10.2.4, 10.2.5
t	rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
📅	timestamp	Oct 25, 2026 @ 12:53:50.278

9. Conclusion:

This project provided practical exposure to Wazuh SIEM, Docker-based deployment, Windows endpoint monitoring, security event review, and basic alert triage.

The Wazuh dashboard displayed Windows endpoint events and authentication-related information, allowing individual event fields to be inspected.

The current lab establishes a foundation for SOC Analyst learning. Future work will focus on validating Sysmon telemetry, investigating selected alerts end-to-end, and documenting reproducible detection scenarios with supporting evidence.

9. Future Scope:

1.Validate Sysmon event collection and selected event IDs.
2.Investigate failed-login activity using Windows Security events.
3.Correlate related events to improve alert triage.
4.Document one complete investigation with evidence and conclusion.
5.Add screenshots and setup instructions to GitHub.
6.Explore additional Wazuh detection and monitoring capabilities